MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – CHAPTER TEST (SET – 3)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The distance of point $P(3a, 4a)$ from y-axis is: [1]

(A) $3a$ **(B)** $-3a$ **(C)** $4a$ **(D)** $-4a$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $3a$

Given: Point $P(3a, 4a)$. Perpendicular foot on the y-axis is $N(0, 4a)$.

Formula Used: Distance of (x, y) from y-axis $= |x|$

$$PN = \sqrt{(0 - 3a)^2 + (4a - 4a)^2} = \sqrt{9a^2} = |3a|$$

Assuming $a > 0$ (as the printed options omit modulus signs), the distance is **3a units**.

Q2. The distance of point $(a, -b)$ from x-axis is: [1]

(A) a **(B)** $-a$ **(C)** b **(D)** $-b$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) b

Given: Point $P(a, -b)$. Foot of perpendicular on the x-axis is $M(a, 0)$.

Formula Used: Distance of (x, y) from x-axis $= |y|$

$$PM = \sqrt{(a - a)^2 + (0 - (-b))^2} = \sqrt{b^2} = |b|$$

Since distance is always non-negative (with $b > 0$), the distance is **b units**.

Q3. The distance of the point $A(-3,-4)$ from x-axis is: [1]

(A) 3 **(B)** 4 **(C)** 5 **(D)** 7

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 4

Given: Point $A(-3,-4)$.

Formula Used: Distance of (x,y) from x-axis $= |y|$

$$\text{Distance} = |-4| = \boxed{\textbf{4 units}}$$

Q4. The point on y-axis equidistant from the points $A(1,3)$ and $B(4,4)$ is: [1]

(A) $(0,11)$ **(B)** $(11,0)$ **(C)** $(0,13)$ **(D)** $(0,12)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $(0,11)$

Given: Let the point on the y-axis be $P(0,y)$, equidistant from $A(1,3)$ and $B(4,4)$, so $PA^2 = PB^2$.

Formula Used: $d^2 = (x_2-x_1)^2 + (y_2-y_1)^2$

$$(0-1)^2 + (y-3)^2 = (0-4)^2 + (y-4)^2$$

$$1 + (y^2 - 6y + 9) = 16 + (y^2 - 8y + 16) \implies y^2 - 6y + 10 = y^2 - 8y + 32$$

$$-6y + 8y = 32 - 10 \implies 2y = 22 \implies y = 11 \implies \boxed{\textbf{(0, 11)}}$$

Q5. The perimeter of a triangle with vertices $(0,4)$, $(0,0)$ and $(3,0)$ is: [1]

(A) 5 units **(B)** 12 units **(C)** 11 units **(D)** $7+\sqrt{5}$ units

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 12 units

Given: Vertices $A(0,4)$, $O(0,0)$, and $B(3,0)$.

Formula Used: $d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

Calculating each side length:

$$OA = 4 \text{ units}, \quad OB = 3 \text{ units}, \quad AB = \sqrt{(3-0)^2 + (0-4)^2} = \sqrt{9+16} = \sqrt{25} = 5 \text{ units}$$

$$\text{Perimeter} = OA + OB + AB = 4 + 3 + 5 = \boxed{\textbf{12 units}}$$

Q6. The distance of a point A from x-axis is 3 units. Which of the following cannot be coordinates of the point A?
(a) $(1, 3)$ (b) $(-3, -3)$ (c) $(-3, 3)$ (d) $(3, 1)$ [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Condition for Distance from x-Axis [1 Mark]
We know that the perpendicular distance of any point $A(x, y)$ from the x-axis is the absolute value of its y-coordinate (ordinate):

Formula Used: Distance of (x, y) from x-axis $= |y|$

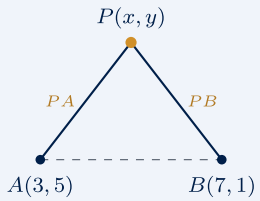
For the distance from the x-axis to be 3 units, we must have $|y| = 3 \implies y = +3$ or $y = -3$.

Step 2: Checking Each Option Step by Step [2 Marks]
- Option (a) $(1, 3)$: $|y| = |3| = 3$ units (Valid).
- Option (b) $(-3, -3)$: $|y| = |-3| = 3$ units (Valid).
- Option (c) $(-3, 3)$: $|y| = |3| = 3$ units (Valid).
- Option (d) $(3, 1)$: $|y| = |1| = 1 \neq 3$ units (Here 3 is the distance from the y-axis, **not** from the x-axis).

Cannot be coordinates of A : **(d) $(3, 1)$**

Q7. Find a relation between x and y such that $P(x, y)$ is equidistant from the points $A(3, 5)$ and $B(7, 1)$. Hence, write the coordinates of the points on x-axis and y-axis which are equidistant from points A and B. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Finding Linear Relation from $PA = PB$ [1.5 Marks]
Given: $P(x, y)$ is equidistant from $A(3, 5)$ and $B(7, 1)$, so $PA = PB \implies PA^2 = PB^2$. Applying the squared distance formula expresses PA^2 and PB^2 algebraically to simplify into a linear relation between x and y.

We know the squared distance formula between (x_1, y_1) and (x_2, y_2):

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

$$(x - 3)^2 + (y - 5)^2 = (x - 7)^2 + (y - 1)^2$$

$$(x^2 - 6x + 9) + (y^2 - 10y + 25) = (x^2 - 14x + 49) + (y^2 - 2y + 1)$$

Cancelling $x^2 + y^2$ from both sides:

$$-6x - 10y + 34 = -14x - 2y + 50$$

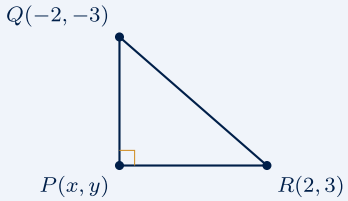
$$-6x + 14x - 10y + 2y = 50 - 34 \implies 8x - 8y = 16 \implies \mathbf{x - y = 2}$$

Step 2: Points on x-Axis and y-Axis Equidistant from A and B [1.5 Marks]
- For the x-axis, put $y = 0$ into $x - y = 2$: $x - 0 = 2 \implies x = 2 \implies \mathbf{(2, 0)}$.
- For the y-axis, put $x = 0$ into $x - y = 2$: $0 - y = 2 \implies y = -2 \implies \mathbf{(0, -2)}$.

Relation: $\mathbf{x - y = 2}$; x-axis point: $\mathbf{(2, 0)}$; y-axis point: $\mathbf{(0, -2)}$

Q8. $P(x, y)$, $Q(-2, -3)$ and $R(2, 3)$ are the vertices of a right triangle PQR right angled at P. Find the relationship between x and y. Hence, find all possible values of x for which $y = 2$. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Step 1: Applying Pythagoras' Theorem at Right Angle P [1 Mark]

Given: $\triangle PQR$ has $\angle P = 90°$ with $P(x, y)$, $Q(-2, -3)$, and $R(2, 3)$. By Pythagoras' theorem, $PQ^2 + PR^2 = QR^2$, which gives the required locus condition.

We know the squared distance formula:

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Step 2: Deriving Relation Between x and y [2.5 Marks]

Computing squared lengths:

$$PQ^2 = (x + 2)^2 + (y + 3)^2 = x^2 + 4x + 4 + y^2 + 6y + 9$$

$$PR^2 = (x - 2)^2 + (y - 3)^2 = x^2 - 4x + 4 + y^2 - 6y + 9$$

$$QR^2 = (2 - (-2))^2 + (3 - (-3))^2 = 4^2 + 6^2 = 16 + 36 = 52$$

Substituting into $PQ^2 + PR^2 = QR^2$:

$$(2x^2 + 2y^2 + 8 + 18) = 52 \implies 2x^2 + 2y^2 + 26 = 52 \implies 2x^2 + 2y^2 = 26 \implies \mathbf{x^2 + y^2 = 13}$$

Step 3: Finding Possible Values of x when $y = 2$ [1.5 Marks]

Substituting $y = 2$ into $x^2 + y^2 = 13$:

$$x^2 + 2^2 = 13 \implies x^2 + 4 = 13 \implies x^2 = 9 \implies x = \pm 3$$

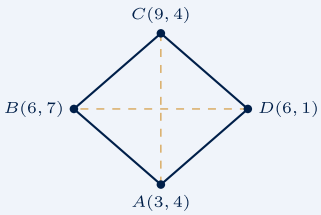
(Both $P(3, 2)$ and $P(-3, 2)$ are distinct from Q and R, forming valid right triangles.)

Relation: $\mathbf{x^2 + y^2 = 13}$	and	When $y = 2$, $\mathbf{x = 3}$ or $\mathbf{x = -3}$

Q9. Competency Case Study: In a classroom, 4 friends are seated at the points A, B, C and D as shown in the Cartesian grid. Champa and Chameli walk into the class and after observing for a few minutes Champa asks Chameli, "Don't you think $ABCD$ is a square?" Chameli disagrees. The coordinates are $A(3,4)$, $B(6,7)$, $C(9,4)$ and $D(6,1)$. [4]

(i) What is the distance AB?
(ii) What is the distance BC?
(iii) What is the length of the diagonal AC?
(iv) Is $ABCD$ a square or a rhombus but not a square?

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Given: Seating coordinates $A(3,4)$, $B(6,7)$, $C(9,4)$, and $D(6,1)$. To identify whether $ABCD$ is a square or a non-square rhombus, we compute adjacent side lengths AB, BC, CD, DA and both diagonals AC and BD.

We know the distance formula:

Formula Used: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

(i) Distance AB: [1 Mark]

$$AB = \sqrt{(6-3)^2 + (7-4)^2} = \sqrt{3^2 + 3^2} = \sqrt{18} = \mathbf{3\sqrt{2}} \text{ units}$$

(ii) Distance BC: [1 Mark]

$$BC = \sqrt{(9-6)^2 + (4-7)^2} = \sqrt{3^2 + (-3)^2} = \sqrt{9+9} = \sqrt{18} = \mathbf{3\sqrt{2}} \text{ units}$$

(iii) Length of Diagonal AC: [1 Mark]

$$AC = \sqrt{(9-3)^2 + (4-4)^2} = \sqrt{6^2 + 0^2} = \sqrt{36} = \mathbf{6} \text{ units}$$

(iv) Checking Square vs Non-Square Rhombus: [1 Mark]

Similarly, computing the remaining sides and second diagonal:

$$CD = \sqrt{(6-9)^2 + (1-4)^2} = \sqrt{9+9} = 3\sqrt{2}, \qquad DA = \sqrt{(3-6)^2 + (4-1)^2} = \sqrt{9+9} = 3\sqrt{2}$$

$$BD = \sqrt{(6-6)^2 + (1-7)^2} = \sqrt{0+36} = \mathbf{6} \text{ units}$$

All four sides are equal ($AB = BC = CD = DA = 3\sqrt{2}$) and both diagonals are equal ($AC = BD = 6$). Also $AB^2 + BC^2 = 18 + 18 = 36 = AC^2 \implies \angle B = 90°$.

$\boxed{AB = 3\sqrt{2},\ BC = 3\sqrt{2},\ AC = 6; \quad ABCD \text{ is a } \textbf{square} \text{ (Champa is correct).}}$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★